Formalizing Non-Equilibrium Stochastic Thermodynamics, Jarzynski–Crooks Fluctuation Theorems, and ATP-Dependent Chromatin Remodeling Lattice Dynamics in Lean 4

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Abstract

ATP-dependent chromatin remodeling complexes—specifically human ISWI (`SNF2H/SMARCA5`, UniProt `O60264`, Cryo-EM structure `PDB: 6NE3` at 3.90 Å) and SWI/SNF (`BRG1/SMARCA4`, UniProt `P51532`, Cryo-EM structure `PDB: 6LTJ` at 3.70 Å)—reposition 147-bp Widom 601 nucleosome core particles (`NCP147`, X-ray structure `PDB: 1KX5` at 1.94 Å) far from thermodynamic equilibrium by coupling discrete DNA translocation cycles at superhelical location 2 (`SHL-2`) to the chemical potential of adenosine triphosphate hydrolysis ($\Delta\mu_{\mathrm{ATP}} = 20\,k_BT = 85.6$ pN · nm at $T = 310$ K). Here we present a complete, axiom-free formalization of continuous-time non-equilibrium stochastic thermodynamics, single-molecule fluctuation theorems, and multi-nucleosome chromatin polymer physics in the Lean 4 theorem prover (`ANSE.StochasticThermodynamics`), leveraging `mathlib4`'s real calculus (`HasDerivAt`), exponential convexity (`Real.add_one_le_exp`), and finite-sum Cauchy–Schwarz machinery. First, for any n-state continuous-time Markov chain governed by the Pauli Master Equation $\frac{df_i}{dt} = \sum_{j=0}^{n-1}(W_{ij}f_j - W_{ji}f_i)$, we prove the universal antisymmetric flux identity $(Wf)_i = \sum_j J_{ij}(f)$, exact total probability conservation $\frac{d}{dt}\sum_i f_i(t) = 0$, and unconditional positivity $f_i^{(k+1)} > 0$ and mass conservation $\sum_i f_i^{(k+1)} = \sum_i f_i^{(k)}$ of Backward-Euler / L-stable Implicit-Explicit (IMEX) time integration $(I - \Delta t W)f^{(k+1)} = f^{(k)}$ for all $\Delta t > 0$ and all $\Delta\mu_{\mathrm{ATP}} \in \mathbb{R}$, contrasting this with stiff explicit Forward-Euler step-size breakdown $1 - \Delta t\, w_{\mathrm{exit}} = -3/4 < 0$ at $\Delta t = 1/4 > 1/7$ (Theorem 2.1, `ST_T1`). Second, we formalize the exact time derivative of Shannon system entropy $S_{\mathrm{sys}}(f(t)) = -\sum_i f_i(t)\ln f_i(t)$ via `HasDerivAt`, prove the exact Schnakenberg decomposition $\dot\Sigma(f) = \dot S_{\mathrm{sys}}(f) + \dot S_{\mathrm{med}}(f)$, and establish the universal Second Law of Non-Equilibrium Biophysics $\dot\Sigma(f) \geq 0$ for all $n \in \mathbb{N}$ and $f_i > 0$ (Theorem 2.2, `ST_T2`). Third, under Local Detailed Balance $\ln(W_{ij}/W_{ji}) = \ln(f_{\mathrm{eq},i}/f_{\mathrm{eq},j}) + m_{ij}\Delta\mu_{\mathrm{ATP}}/(k_BT)$, we formalize the continuous-time Kullback–Leibler divergence derivative and prove the exact Esposito–Van den Broeck non-equilibrium free-energy work balance $k_BT\frac{d}{dt}D_{\mathrm{KL}}(f(t)\|f_{\mathrm{eq}}) = \dot W_{\mathrm{ATP}}(f(t)) - k_BT\dot\Sigma(f(t))$ (Theorem 2.3, `ST_T3`). Fourth, we prove the universal logarithmic–hyperbolic inequality $\frac{2(x-y)^2}{x+y} \leq (x-y)\ln(x/y)$ for all $x, y > 0$ via derivative monotonicity and lift it via Cauchy–Schwarz to the finite-state Barato–Seifert–Gingrich Thermodynamic Uncertainty Relation (TUR) $2J_d(f)^2 \leq \mathcal{A}_{\mathrm{tot}}(f)\dot\Sigma(f)$ (Theorem 2.4, `ST_T4`). Fifth, we formally verify the Crooks Fluctuation Theorem $\ln(P_F(\gamma)/P_R(\gamma)) = \beta(W(\gamma) - \Delta F)$, the Jarzynski Equality $\langle e^{-\beta W}\rangle = e^{-\beta\Delta F}$, the Second-Law dissipated-work inequality $\Delta F \leq \langle W\rangle$, and Schnakenberg's macroscopic cycle entropy production rate $\dot\Sigma_{\mathrm{cycle}} = J_{\mathrm{cyc}}\Delta\mu_{\mathrm{ATP}}/(k_BT) \geq 0$ (Theorem 2.5, `ST_T5`). Sixth, we scale to multi-nucleosome chromatin state spaces by formalizing the 1D McGhee–von Hippel steric exclusion factor $g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}}, \rho) = \frac{1-\ell_{\mathrm{nuc}}\rho}{1-(\ell_{\mathrm{nuc}}-1)\rho} \in (0, 1)$ for $\ell_{\mathrm{nuc}} = 147$ bp, its crowding derivative $\partial_\rho g_{\mathrm{MvH}} = -1/(1-(\ell_{\mathrm{nuc}}-1)\rho)^2 < 0$, the 1D Tonks–Girardeau hard-rod osmotic pressure $\Pi_{\mathrm{TG}}/(k_BT) = \rho/(1-\ell_{\mathrm{nuc}}\rho)$ ($53/18 \approx 2.94\times$ pressure surge from 200 bp euchromatin to 165 bp heterochromatin), continuous 3D Rouse bead-spring chromatin polymer Langevin dissipation $\frac{d}{dt}U_{\mathrm{Rouse}} = -\zeta_{\mathrm{nuc}}^{-1}\sum_i G_i^2 \leq 0$, and empirical single-molecule FRET (smFRET) and optical-tweezers unwrapping observables ($\Delta G^{\ddagger}_{\mathrm{unwrap}} = 6.5\,k_BT = 27.82$ pN · nm, $D_{\mathrm{slide}} = 0.36$ bp^2/s, $k_{\mathrm{cat}} = 2.4$ bp/s, Theorem 2.6, `ST_T6`). All six theorems compile in Lean 4 with zero `sorryAx` and zero custom axioms.

1 Introduction: ATP-Dependent Chromatin Remodeling and Epigenetic Polymer Physics

Eukaryotic genomes package 147 bp of B-form DNA in 1.65 left-handed superhelical turns around a histone octamer ($(H3–H4)_2(H2A–H2B)_2$) to form the canonical nucleosome core particle (`NCP147`, X-ray crystal structure `PDB: 1KX5` at 1.94 Å resolution, 200.28 kDa) [16, 6]. Because thermal fluctuations alone induce only slow passive nucleosome diffusion ($D_{\text{slide}} \approx 0.36$ bp^2/s $= 0.0416$ nm^2/s) and face a substantial outer-turn 20-bp DNA unwrapping barrier $\Delta G^{\ddagger}_{\text{unwrap}} \approx 6.5\, k_BT = 27.82$ pN $\cdot$ nm measured by single-molecule optical tweezers [11], eukaryotic cells rely on specialized superfamily-2 (SF2) ATP-dependent chromatin remodeling motors to regulate DNA accessibility during transcription, replication, and repair [4].

Specifically, human **ISWI** (`SNF2H/SMARCA5`, UniProt `O60264`, 1052 AA, Cryo-EM complex bound to the nucleosome at superhelical location 2 (`SHL-2`), `PDB: 6NE3` at 3.90 Å resolution [1]) and human **SWI/SNF** (`BRG1/SMARCA4`, UniProt `P51532`, 1647 AA, Cryo-EM nucleosome-bound BAF complex, `PDB: 6LTJ` at 3.70 Å resolution [12]) transduce the physiological chemical free energy of ATP hydrolysis (at $T = 310$ K, $k_BT = 4.28$ pN $\cdot$ nm)

$$\Delta\mu_{\text{ATP}} = \mu_{\text{ATP}} - \mu_{\text{ADP}} - \mu_{\text{P}_\text{i}} = 20\, k_BT = 85.6 \text{ pN} \cdot \text{nm} \approx 51.6 \text{ kJ/mol} \tag{1}$$

into directed 1-bp (0.34 nm) DNA translocation steps at `SHL-2` that accumulate into 3–7 bp (1.02–2.38 nm) remodeling bursts at catalytic velocity $k_{\text{cat}} \approx 2.4$ bp/s $= 0.816$ nm/s, as directly resolved by single-molecule Förster resonance energy transfer (smFRET) trajectories [3, 7].

Across this chromatin remodeling hierarchy, the probability distribution $f(t) = (f_0(t), \ldots, f_{n-1}(t)) \in \Delta^{n-1}$ over n discrete nucleosome-remodeler microstates is governed by the continuous-time **Pauli Master Equation** [18, 14, 19]

$$\frac{df_i(t)}{dt} = \sum_{j=0}^{n-1} (W_{ij} f_j(t) - W_{ji} f_i(t)) = \sum_{j=0}^{n-1} J_{ij}(f(t)), \tag{2}$$

where $W_{ij} > 0$ $(i \neq j)$ denotes the transition rate from microstate j to microstate i, $W_{ii} = -\sum_{j\neq i} W_{ji}$ is the exit rate from microstate i, and $J_{ij}(f) = W_{ij} f_j - W_{ji} f_i = -J_{ji}(f)$ is the directed probability flux along transition $j \to i$. When transition $j \to i$ is coupled to the hydrolysis of $m_{ij} = -m_{ji} \in \mathbb{R}$ ATP molecules against a thermal bath at temperature $T = 310$ K, microscopic reversibility imposes **Local Detailed Balance (LDB)** [19, 9]:

$$\ln\left(\frac{W_{ij}}{W_{ji}}\right) = \frac{E_j - E_i + m_{ij}\Delta\mu_{\text{ATP}}}{k_BT} = \ln\left(\frac{f_{\text{eq},i}}{f_{\text{eq},j}}\right) + \frac{m_{ij}\Delta\mu_{\text{ATP}}}{k_BT}, \tag{3}$$

where $f_{\text{eq},i} = Z^{-1}\exp(-E_i/(k_BT))$ is the unforced Boltzmann–Gibbs equilibrium distribution at $\Delta\mu_{\text{ATP}} = 0$.

Rigorous computational and mathematical modeling of ATP-dependent chromatin remodeling requires uniting three physical scales within a single machine-checked framework:

1. **Positivity-Preserving Master-Equation Integration and Network Thermodynamics:** Because ATP chemical potential ($\Delta\mu_{\text{ATP}} = 20\, k_BT$) tilts transition barriers exponentially via Local Detailed Balance, explicit Forward-Euler time stepping $f_i^{(k+1)} = (1 - \Delta t\, w_{\text{exit},i}) f_i^{(k)} + \Delta t \sum_{j\neq i} W_{ij} f_j^{(k)}$ violates probability positivity whenever $\Delta t > 1/w_{\text{exit},i}$, whereas Backward-Euler and L-stable Implicit-Explicit (IMEX) schemes preserve $f_i^{(k+1)} > 0$ and $\sum_i f_i^{(k+1)} = 1$ unconditionally for all $\Delta t > 0$ while obeying Schnakenberg's network entropy decomposition [18], the Esposito–Van den Broeck free-energy balance [9], and the Barato–Seifert–Gingrich Thermodynamic Uncertainty Relation (TUR) [2, 10].

2. **Single-Molecule Non-Equilibrium Fluctuation Theorems (Crooks and Jarzynski):** When an optical trap or an ATP-driven remodeler (`SNF2H PDB: 6NE3` or `BRG1 PDB: 6LTJ`) pulls the outer DNA turn off the histone octamer along a non-equilibrium trajectory γ, the forward and time-reversed path ensembles $P_F(\gamma)$ and $P_R(\gamma)$ obey the Crooks Fluctuation Theorem [5] and the Jarzynski Equality [13], recovering the equilibrium unwrapping free-energy barrier $\Delta F_{\text{unwrap}} = 6.5\, k_B T = 27.82$ pN $\cdot$ nm from non-equilibrium work measurements $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$.

3. **Multi-Nucleosome 1D Lattice Gas Steric Exclusion and 3D Chromatin Polymer Dynamics:** Along a genomic DNA fiber of nucleosome repeat length NRL $= 1/\rho$, each nucleosome occupies an excluded footprint of $\ell_{\text{nuc}} = 147$ bp (`PDB: 1KX5`). Multi-nucleosome sliding is governed by the 1D McGhee–von Hippel steric accessibility factor [17] and Tonks–Girardeau hard-rod osmotic pressure [20, 15], coupled to 3D Rouse bead-spring chromatin polymer relaxation [8].

In this paper, we present a 100% self-contained, axiom-free formalization of all six theorems (`ST_T1`–`ST_T6`) in Lean 4 (`formal/ANSE/StochasticThermodynamicsSpec.lean` and `formal/ANSE/StochasticThermodynamics.lean`), coupled with deterministic numerical execution in `rusty-SUNDIALS` (`ChromatinRemodelerLatticeEngine` in `crates/sundials-core/src/genetics.rs`). Table 1 stratifies the six theorems across Formal Kernel proofs in Lean 4, Exact Rational thermodynamic and steric certificates, and Validated Numerical Benchmarks.

Table 1: Summary of Lean 4 theorems (`ANSE.StochasticThermodynamics`) and certified biophysical invariants for ATP-dependent chromatin remodeling (`SNF2H PDB: 6NE3`, `BRG1 PDB: 6LTJ`, `NCP147 PDB: 1KX5`).

Result	**Lean 4 Theorem & Mechanism**	**Certified Biophysical Law / Experimental Invariant**
Theorem 2.1	`ST_T1`: Pauli Master Eq. & IMEX Positivity	$\sum_i (Wf)_i = 0$, $(I - \Delta t W) f^{(k+1)} = f^{(k)} \implies f_i^{(k+1)} > 0$, CFL breakdown $1 - \Delta t\, w_{\text{exit}} < 0$
Theorem 2.2	`ST_T2`: Schnakenberg Decomposition & 2nd Law	$\frac{d}{dt} S_{\text{sys}} + \dot{S}_{\text{med}} = \dot{\Sigma}(f) \geq 0$ $(\forall n \in \mathbb{N},\ f_i > 0)$
Theorem 2.3	`ST_T3`: Esposito–Van den Broeck Balance	$k_B T \frac{d}{dt} D_{\text{KL}}(f \| f_{\text{eq}}) = \dot{W}_{\text{ATP}}(f) - k_B T \dot{\Sigma}(f)$
Theorem 2.4	`ST_T4`: Thermodynamic Uncertainty Relation	$\frac{2(x-y)^2}{x+y} \leq (x-y)\ln(x/y) \implies 2J_d(f)^2 \leq \mathcal{A}_{\text{tot}}(f)\dot{\Sigma}(f)$
Theorem 2.5	`ST_T5`: Crooks, Jarzynski & Cycle Affinity	$\ln \frac{P_F(\gamma)}{P_R(\gamma)} = \beta(W(\gamma) - \Delta F)$, $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$, $\Delta F \leq \langle W \rangle$, $\dot{\Sigma}_{\text{cyc}} = J_{\text{cyc}} \frac{\Delta \mu_{\text{ATP}}}{k_B T} \geq 0$
Theorem 2.6	`ST_T6`: McGhee–von Hippel, Tonks & Rouse	$g_{\text{MvH}}(147, \rho) \in (0,1)$ $(53/54 \to 18/19)$, Π_{TG} surge $53/18 \approx 2.94\times$, $\dot{U}_{\text{Rouse}} = -\zeta^{-1} \sum_i G_i^2 \leq 0$, $\Delta G^{\ddagger}_{\text{unwrap}} = 27.82$ pN $\cdot$ nm

2 Main Theorems and Analytical Demonstrations

2.1 Pauli Master Equation Flux Identity, Probability Mass Conservation, and Unconditional Backward-Euler / IMEX Positivity

Let `ConformationalNetwork n` be an n-state continuous-time Markov chain modeling the discrete mechanochemical states of an ATP-dependent chromatin remodeler with transition rate matrix $W : \text{Fin}\, n \to \text{Fin}\, n \to \mathbb{R}$ satisfying strict off-diagonal positivity $W_{ij} > 0$ $(i \neq j)$ and exact column-sum conservation $\sum_{i=0}^{n-1} W_{ij} = 0$ for every column $j \in \text{Fin}\, n$ (so that $W_{jj} = -\sum_{i \neq j} W_{ij} < 0$).

Theorem 2.1 (Pauli Master Equation Flux Decomposition, Mass Conservation, and Unconditional Backward-Euler / L-Stable IMEX Positivity)**.** *Formalized in Lean 4 as* `ST_T1_master_equation_mass_conservation_and_imex_positivity`*.*

1. ***Antisymmetric Edge-Flux Decomposition and Universal Mass Conservation:*** *For every state vector $f : \mathrm{Fin}\, n \to \mathbb{R}$, the Pauli Master Equation operator $(Wf)_i = \sum_{j=0}^{n-1} W_{ij} f_j$ decomposes into pairwise antisymmetric edge fluxes $J_{ij}(f) = W_{ij} f_j - W_{ji} f_i = -J_{ji}(f)$ as*

$$(Wf)_i = \sum_{j=0}^{n-1} J_{ij}(f), \qquad \sum_{i=0}^{n-1} (Wf)_i = 0. \tag{4}$$

   Consequently, along any differentiable solution $\frac{df_i(t)}{dt} = (Wf(t))_i$, total probability mass is conserved identically: $\frac{d}{dt} \sum_{i=0}^{n-1} f_i(t) = 0$.

2. ***Unconditional Backward-Euler / L-Stable IMEX Mass Conservation and Positivity:*** *Let $\Delta t > 0$ be any time step and let $f^{(k)}, f^{(k+1)} : \mathrm{Fin}\, n \to \mathbb{R}$ satisfy the implicit Backward-Euler (or L-stable IMEX stiff stage) linear system*

$$f_i^{(k+1)} - \Delta t\, (W f^{(k+1)})_i = f_i^{(k)} \qquad (\forall i \in \mathrm{Fin}\, n). \tag{5}$$

   Then: (a) $\sum_{i=0}^{n-1} f_i^{(k+1)} = \sum_{i=0}^{n-1} f_i^{(k)}$ for all $\Delta t > 0$; and (b) for any non-empty network ($n > 0$) possessing a strictly positive stationary distribution $\pi_i > 0$ ($W\pi = 0$), strict positivity of the prior step ($f_i^{(k)} > 0$ for all i) implies strict positivity of the updated step ($f_i^{(k+1)} > 0$ for all i) unconditionally for all $\Delta t > 0$ and all ATP chemical potentials $\Delta\mu_{\mathrm{ATP}} \in \mathbb{R}$.

3. ***Forward-Euler Stiff CFL Positivity Breakdown:*** *Under explicit Forward-Euler integration from vertex state $f = e_j$, the diagonal population multiplier $1 - \Delta t\, w_{\mathrm{exit},j}$ becomes strictly negative if and only if $\Delta t > 1/w_{\mathrm{exit},j}$. In particular, for a 3-state ATP-driven remodeling cycle with exit rate $w_{\mathrm{exit}} = 6 + 1 = 7\ \mathrm{s}^{-1}$ at step size $\Delta t = 1/4\ \mathrm{s} > 1/7\ \mathrm{s}$, the diagonal multiplier evaluates to $1 - (1/4)(7) = -3/4 < 0$.*

Proof. In part (i), expanding $\sum_{j=0}^{n-1}(W_{ij} f_j - W_{ji} f_i) = \sum_{j=0}^{n-1} W_{ij} f_j - f_i \sum_{j=0}^{n-1} W_{ji}$ and applying $\sum_{j=0}^{n-1} W_{ji} = 0$ yields $(Wf)_i = \sum_j J_{ij}(f)$. Swapping the finite double sum via `Finset.sum_comm` gives $\sum_i \sum_j W_{ij} f_j = \sum_j f_j \left(\sum_i W_{ij}\right) = 0$, and linearity of differentiation (`HasDerivAt.fun_sum`) gives $\frac{d}{dt} \sum_i f_i(t) = \sum_i (Wf(t))_i = 0$.

In part (ii), summing $f_i^{(k+1)} - \Delta t (W f^{(k+1)})_i = f_i^{(k)}$ over i and using $\sum_i (W f^{(k+1)})_i = 0$ proves $\sum_i f_i^{(k+1)} = \sum_i f_i^{(k)}$. To prove strict positivity $f_i^{(k+1)} > 0$ without matrix inversion assumptions, define the normalized state ratio $r_i = f_i^{(k+1)}/\pi_i$ so that $f_i^{(k+1)} = \pi_i r_i$. Since $\mathrm{Fin}\, n$ is finite and non-empty, `Finset.exists_min_image` furnishes an index $i_0 \in \mathrm{Fin}\, n$ minimizing $r_{i_0} \le r_j$ for all $j \in \mathrm{Fin}\, n$. Using $W\pi = 0$, we expand $(W f^{(k+1)})_{i_0}$ as

$$(W f^{(k+1)})_{i_0} = \sum_{j=0}^{n-1} W_{i_0 j} \pi_j r_j - r_{i_0} \sum_{j=0}^{n-1} W_{i_0 j} \pi_j = \sum_{j=0}^{n-1} W_{i_0 j} \pi_j (r_j - r_{i_0}).$$

For $j = i_0$, the summand vanishes ($r_{i_0} - r_{i_0} = 0$); for $j \neq i_0$, $W_{i_0 j} > 0$, $\pi_j > 0$, and $r_j - r_{i_0} \ge 0$, so every term is non-negative and $(W f^{(k+1)})_{i_0} \ge 0$. Substituting into the Backward-Euler equation at i_0 with $\Delta t > 0$ yields

$$\pi_{i_0} r_{i_0} = f_{i_0}^{(k+1)} = f_{i_0}^{(k)} + \Delta t\, (W f^{(k+1)})_{i_0} \ge f_{i_0}^{(k)} > 0,$$

whence $r_{i_0} > 0$. Then for every $i \in \mathrm{Fin}\, n$, $r_i \ge r_{i_0} > 0$, so $f_i^{(k+1)} = \pi_i r_i > 0$. Part (iii) follows from $1 - \Delta t\, w_{\mathrm{exit}} < 0 \iff 1 < \Delta t\, w_{\mathrm{exit}} \iff 1/w_{\mathrm{exit}} < \Delta t$ and $1 - (1/4)(7) = -3/4 < 0$. □

2.2 Continuous-Time Shannon Entropy Derivative, Exact Schnakenberg Decomposition, and the Second Law of Non-Equilibrium Biophysics

For a strictly positive microstate distribution $f_i > 0$, the system Shannon entropy (in units of k_B) is $S_{\text{sys}}(f) = -\sum_{i=0}^{n-1} f_i \ln f_i$, the environmental medium entropy flux is $\dot{S}_{\text{med}}(f) = \frac{1}{2}\sum_{i,j} J_{ij}(f)\ln(W_{ij}/W_{ji})$, and Schnakenberg's total entropy production rate [18] is

$$\dot{\Sigma}(f) = \frac{1}{2}\sum_{i=0}^{n-1}\sum_{j=0}^{n-1}(W_{ij}f_j - W_{ji}f_i)\ln\left(\frac{W_{ij}f_j}{W_{ji}f_i}\right). \tag{6}$$

Theorem 2.2 (Continuous-Time Shannon Entropy Derivative, Exact Schnakenberg Decomposition, and Universal Second Law)**.** *Formalized in Lean 4 as* `ST_T2_schnakenberg_entropy_decomposition_and_second_law`*.*

1. ***Universal Two-Point Convex Log-Sum Inequality:*** *For all real forward and reverse transition fluxes $x, y > 0$,*
$$0 \le (x-y)\ln\left(\frac{x}{y}\right). \tag{7}$$

2. ***Continuous-Time Shannon Entropy Derivative (`HasDerivAt`):*** *Along any differentiable trajectory $f : \mathbb{R} \to \operatorname{Fin} n \to \mathbb{R}$ satisfying $f_i(t) > 0$ and `HasDerivAt` $(s \mapsto f_i(s))$ $((Wf(t))_i)$ t for all $i \in \operatorname{Fin} n$, the Shannon entropy $S_{\text{sys}}(f(s)) = -\sum_{i=0}^{n-1} f_i(s)\ln f_i(s)$ has exact time derivative*
$$\frac{d}{dt}S_{\text{sys}}(f(t)) = -\sum_{i=0}^{n-1}(Wf(t))_i\,(\ln f_i(t) + 1) = -\sum_{i=0}^{n-1}(Wf(t))_i \ln f_i(t). \tag{8}$$

3. ***Exact Schnakenberg Decomposition and Universal Second Law:*** *For every $n \in \mathbb{N}$ and every strictly positive state distribution $f_i > 0$, total entropy production splits identically into system and medium entropy rates and is universally non-negative:*
$$\dot{\Sigma}(f) = \frac{d}{dt}S_{\text{sys}}(f) + \dot{S}_{\text{med}}(f), \qquad \dot{\Sigma}(f) \ge 0. \tag{9}$$

Proof. For part (i), if $y \le x$, then $x - y \ge 0$ and $x/y \ge 1$, so $\ln(x/y) \ge 0$ (`Real.log_nonneg`) and $(x-y)\ln(x/y) \ge 0$; if $x < y$, symmetry via $\ln(x/y) = -\ln(y/x)$ reduces to the first case. In part (ii), applying the product and chain rules (`HasDerivAt.mul` and `HasDerivAt.log`) to $s \mapsto f_i(s)\ln f_i(s)$ gives derivative $(Wf(t))_i \ln f_i(t) + f_i(t)\frac{(Wf(t))_i}{f_i(t)} = (Wf(t))_i(\ln f_i(t) + 1)$, and summing over $i \in \operatorname{Fin} n$ with $\sum_i (Wf(t))_i = 0$ eliminates the $+1$ constant. In part (iii), for $i \ne j$, strict positivity $W_{ij}, W_{ji}, f_i, f_j > 0$ allows splitting the logarithm via `Real.log_mul` and `Real.log_div`:

$$J_{ij}(f)\ln\left(\frac{W_{ij}f_j}{W_{ji}f_i}\right) = J_{ij}(f)\ln\left(\frac{W_{ij}}{W_{ji}}\right) + J_{ij}(f)\,(\ln f_j - \ln f_i).$$

By the antisymmetry $J_{ji}(f) = -J_{ij}(f)$ and `Finset.sum_comm`, the bilinear double sum satisfies $\frac{1}{2}\sum_{i,j} J_{ij}(f)(g_j - g_i) = -\sum_i \left(\sum_j J_{ij}(f)\right) g_i = -\sum_i (Wf)_i g_i$. Setting $g_i = \ln f_i$ yields $\frac{1}{2}\sum_{i,j} J_{ij}(f)(\ln f_j - \ln f_i) = -\sum_i (Wf)_i \ln f_i = \dot{S}_{\text{sys}}(f)$, proving $\dot{\Sigma}(f) = \dot{S}_{\text{sys}}(f) + \dot{S}_{\text{med}}(f)$. Finally, applying part (i) to $x = W_{ij}f_j > 0$ and $y = W_{ji}f_i > 0$ for each $i \ne j$ proves $\dot{\Sigma}(f) \ge 0$. □

2.3 Local Detailed Balance and the Exact Esposito–Van den Broeck Non-Equilibrium Free-Energy Work Balance

Let $f_{\mathrm{eq},i} > 0$ be the reference unforced Boltzmann distribution and let $D_{\mathrm{KL}}(f\|f_{\mathrm{eq}}) = \sum_{i=0}^{n-1} f_i \ln(f_i/f_{\mathrm{eq},i})$ denote the relative Kullback–Leibler divergence (so that $k_B T\, D_{\mathrm{KL}}(f\|f_{\mathrm{eq}}) = F(f) - F_{\mathrm{eq}}$ is the excess non-equilibrium Helmholtz free energy). When transition $j \to i$ hydrolyzes $m_{ij} = -m_{ji}$ ATP molecules at chemical potential $\Delta\mu_{\mathrm{ATP}}$, the rate of chemical work injected by ATP hydrolysis is

$$\dot{W}_{\mathrm{ATP}}(f) = \frac{1}{2}\sum_{i=0}^{n-1}\sum_{j=0}^{n-1} J_{ij}(f)\, m_{ij}\, \Delta\mu_{\mathrm{ATP}}. \tag{10}$$

Theorem 2.3 (Continuous-Time Kullback–Leibler Derivative and Exact Esposito–Van den Broeck Free-Energy Work Balance)**.** *Formalized in Lean 4 as* `ST_T3_esposito_van_den_broeck_free_energy_balance`*.*

1. ***Continuous-Time Kullback–Leibler Derivative (`HasDerivAt`):*** *Along any differentiable solution of the Pauli Master Equation with $f_i(t) > 0$ and $f_{\mathrm{eq},i} > 0$,*

$$\frac{d}{dt} D_{\mathrm{KL}}(f(t)\|f_{\mathrm{eq}}) = \sum_{i=0}^{n-1} (Wf(t))_i \left(\ln\left(\frac{f_i(t)}{f_{\mathrm{eq},i}}\right) + 1\right) = \sum_{i=0}^{n-1} (Wf(t))_i \ln\left(\frac{f_i(t)}{f_{\mathrm{eq},i}}\right). \tag{11}$$

2. ***Exact Esposito–Van den Broeck Free-Energy Work Balance:*** *Under Local Detailed Balance $\ln(W_{ij}/W_{ji}) = \ln(f_{\mathrm{eq},i}/f_{\mathrm{eq},j}) + \frac{m_{ij}\Delta\mu_{\mathrm{ATP}}}{k_B T}$ ($i \neq j$, $k_B T > 0$), the excess free-energy rate satisfies the exact identity*

$$k_B T \frac{d}{dt} D_{\mathrm{KL}}(f(t)\|f_{\mathrm{eq}}) = \dot{W}_{\mathrm{ATP}}(f(t)) - k_B T\, \dot{\Sigma}(f(t)). \tag{12}$$

3. ***Equilibrium Lyapunov Decay vs. Non-Equilibrium Steady-State Housekeeping Power:***
   - *At chemical equilibrium ($\Delta\mu_{\mathrm{ATP}} = 0$), $\dot{W}_{\mathrm{ATP}}(f) = 0$ and $\frac{d}{dt} D_{\mathrm{KL}}(f(t)\|f_{\mathrm{eq}}) = -\dot{\Sigma}(f(t)) \leq 0$, proving that $D_{\mathrm{KL}}(\cdot\|f_{\mathrm{eq}})$ is a strict Lyapunov function.*
   - *At any non-equilibrium steady state ($Wf^* = 0$), $\frac{d}{dt} D_{\mathrm{KL}}(f^*\|f_{\mathrm{eq}}) = 0$ and $\dot{W}_{\mathrm{ATP}}(f^*) = k_B T\, \dot{\Sigma}(f^*) \geq 0$, which is strictly positive ($\dot{W}_{\mathrm{ATP}}(f^*) > 0$) whenever $\dot{\Sigma}(f^*) > 0$.*

Proof. Part (i) follows from `HasDerivAt.div_const`, `HasDerivAt.log`, `HasDerivAt.mul`, and `HasDerivAt.fun_sum`, with $\sum_i (Wf)_i = 0$ eliminating the $+1$ term. For part (ii), substituting the Local Detailed Balance relation into the medium entropy rate $\dot{S}_{\mathrm{med}}(f) = \frac{1}{2}\sum_{i,j} J_{ij}(f) \ln(W_{ij}/W_{ji})$ gives

$$\dot{S}_{\mathrm{med}}(f) = \frac{1}{2}\sum_{i,j} J_{ij}(f)\,(\ln f_{\mathrm{eq},i} - \ln f_{\mathrm{eq},j}) + \frac{1}{k_B T}\dot{W}_{\mathrm{ATP}}(f).$$

Using the antisymmetric double-sum lemma $\frac{1}{2}\sum_{i,j} J_{ij}(f)(\ln f_{\mathrm{eq},i} - \ln f_{\mathrm{eq},j}) = \sum_i (Wf)_i \ln f_{\mathrm{eq},i}$, combining with $\dot{S}_{\mathrm{sys}}(f) = -\sum_i (Wf)_i \ln f_i$, and invoking Theorem 2.2(iii) yields

$$\dot{\Sigma}(f) = \dot{S}_{\mathrm{sys}}(f) + \dot{S}_{\mathrm{med}}(f) = -\sum_{i=0}^{n-1} (Wf)_i \ln\left(\frac{f_i}{f_{\mathrm{eq},i}}\right) + \frac{\dot{W}_{\mathrm{ATP}}(f)}{k_B T} = -\frac{d}{dt} D_{\mathrm{KL}}(f\|f_{\mathrm{eq}}) + \frac{\dot{W}_{\mathrm{ATP}}(f)}{k_B T}.$$

Multiplying by $k_B T > 0$ and rearranging proves $k_B T \frac{d}{dt} D_{\mathrm{KL}}(f\|f_{\mathrm{eq}}) = \dot{W}_{\mathrm{ATP}}(f) - k_B T \dot{\Sigma}(f)$. Part (iii) follows immediately by substituting $\Delta\mu_{\mathrm{ATP}} = 0$ (with $\dot{\Sigma}(f) \geq 0$) and $Wf^* = 0$, respectively. □

2.4 Finite-State Thermodynamic Uncertainty Relation (TUR) and Remodeling Activity Bound

For each transition pair (i, j), let $a_{ij}(f) = W_{ij} f_j + W_{ji} f_i \geq 0$ denote the symmetric dynamical traffic (edge activity) and let $\mathcal{A}_{\mathrm{tot}}(f) = \frac{1}{2} \sum_{i,j} a_{ij}(f)$ be the total remodeling activity rate. For any antisymmetric generalized current weight matrix $d_{ij} = -d_{ji}$ with normalized step increments $|d_{ij}| \leq 1$, the generalized net current is $J_d(f) = \frac{1}{2} \sum_{i,j} d_{ij} J_{ij}(f)$.

Theorem 2.4 (Universal Logarithmic–Hyperbolic Inequality and Finite-State Thermodynamic Uncertainty Relation)**.** *Formalized in Lean 4 as* `ST_T4_thermodynamic_uncertainty_relation_and_activity_bound`*.*

1. ***Universal Logarithmic–Hyperbolic Mean Inequality:*** *For all real transition fluxes $x, y > 0$,*
$$\frac{2(x-y)^2}{x+y} \leq (x-y) \ln\left(\frac{x}{y}\right). \tag{13}$$

2. ***Per-Edge Activity–Dissipation Bound:*** *For every state vector $f_i > 0$ and every edge $i \neq j$,*
$$2\, J_{ij}(f)^2 \leq a_{ij}(f)\ (W_{ij} f_j - W_{ji} f_i) \ln\left(\frac{W_{ij} f_j}{W_{ji} f_i}\right). \tag{14}$$

3. ***Global Finite-State Thermodynamic Uncertainty Relation (TUR):*** *For every $n \in \mathbb{N}$, every positive state vector $f_i > 0$, and every generalized current $J_d(f) = \frac{1}{2} \sum_{i,j} d_{ij} J_{ij}(f)$ with $|d_{ij}| \leq 1$,*
$$2\, J_d(f)^2 \leq \mathcal{A}_{\mathrm{tot}}(f)\, \dot{\Sigma}(f). \tag{15}$$

   In particular, for the inhabited 3-state ATP-driven remodeling cycle (`canonicalATPConformationalCycleWitness`*) with forward rate $w_+ = 6\ \mathrm{s}^{-1}$, reverse rate $w_- = 1\ \mathrm{s}^{-1}$, and uniform NESS $\pi = (1/3, 1/3, 1/3)$, the cycle current $J_{\mathrm{cycle}}(\pi) = 5/3\ \mathrm{s}^{-1}$ and total activity $\mathcal{A}_{\mathrm{tot}}(\pi) = 7\ \mathrm{s}^{-1}$ satisfy $2J_{\mathrm{cycle}}(\pi)^2 = 50/9$, rational TUR prefactor $\frac{2J_{\mathrm{cycle}}(\pi)^2}{\mathcal{A}_{\mathrm{tot}}(\pi)(w_+ - w_-)} = 10/63$, and TUR precision–dissipation ratio $\eta_{\mathrm{TUR}}(\pi) = \frac{10}{21 \ln 6} \approx 0.265768 < 1$.*

Proof. To prove part (i) in Lean 4, define the auxiliary function $\psi(u) = u - 2 + \frac{4}{e^u + 1}$ for $u \geq 0$. Using `HasDerivAt.exp` and the quotient rule, ψ has exact derivative

$$\psi'(u) = 1 - \frac{4e^u}{(e^u+1)^2} = \frac{(e^u-1)^2}{(e^u+1)^2} \geq 0 \qquad (\forall u \in \mathbb{R}).$$

By Mathlib's mean-value monotonicity theorem `monotone_of_hasDerivAt_nonneg`, ψ is monotone non-decreasing on $\mathbb{R}$, so $\psi(u) \geq \psi(0) = 0$ for all $u \geq 0$, which rearranges to $\frac{2(e^u-1)}{e^u+1} \leq u$. For $y \leq x$, setting $u = \ln(x/y) \geq 0$ gives $e^u = x/y$, so $\frac{2(x-y)}{x+y} \leq \ln(x/y)$, and multiplying by $x - y \geq 0$ yields $\frac{2(x-y)^2}{x+y} \leq (x-y)\ln(x/y)$; the case $x < y$ follows by symmetry of both sides under $(x, y) \leftrightarrow (y, x)$.

Part (ii) follows by multiplying part (i) at $x = W_{ij} f_j > 0$ and $y = W_{ji} f_i > 0$ by $x + y = a_{ij}(f) > 0$. For part (iii), applying the triangle inequality and $|d_{ij}| \leq 1$ gives $|J_d(f)| \leq \frac{1}{2} \sum_{i,j} |J_{ij}(f)|$. Writing $|J_{ij}(f)| = \sqrt{a_{ij}(f)} \cdot \frac{|J_{ij}(f)|}{\sqrt{a_{ij}(f)}}$ (with $0/0 = 0$ on the diagonal $i = j$) and applying the finite Cauchy–Schwarz inequality (`Finset.sum_mul_sq_le_sq_mul_sq`) over

$\mathrm{Fin}\, n \times \mathrm{Fin}\, n$ yields

$$\left(\sum_{i,j} |J_{ij}(f)|\right)^2 \le \left(\sum_{i,j} a_{ij}(f)\right)\left(\sum_{i,j} \frac{J_{ij}(f)^2}{a_{ij}(f)}\right)$$
$$\le (2\,\mathcal{A}_{\mathrm{tot}}(f))\left(\frac{1}{2}\sum_{i,j} J_{ij}(f)\ln\left(\frac{W_{ij}f_j}{W_{ji}f_i}\right)\right) = 2\,\mathcal{A}_{\mathrm{tot}}(f)\,\dot{\Sigma}(f).$$

Dividing by 2 proves $2J_d(f)^2 \le \mathcal{A}_{\mathrm{tot}}(f)\dot{\Sigma}(f)$. □

2.5 Crooks Fluctuation Theorem, Jarzynski Equality, and Schnakenberg Macroscopic Cycle Affinity Decomposition

In single-molecule optical-tweezers and smFRET experiments on Widom 601 nucleosomes (`NCP147`, `PDB: 1KX5`) remodeled by `ISWI/SNF2H` (`PDB: 6NE3`) or `SWI/SNF` (`PDB: 6LTJ`), an external mechanical force or ATP-driven motor pulls the outer DNA turn across a finite ensemble of K non-equilibrium trajectories $\gamma \in \mathrm{Fin}\, K$ with forward path probabilities $P_F(\gamma)$, time-reversed path probabilities $P_R(\gamma) > 0$, trajectory work $W(\gamma)$, inverse temperature $\beta = 1/(k_B T) > 0$, and equilibrium free-energy difference ΔF [13, 5, 11].

Theorem 2.5 (Crooks Fluctuation Theorem, Jarzynski Equality, Second-Law Work Dissipation, and Schnakenberg Cycle Affinity)**.** *Formalized in Lean 4 as* `ST_T5_jarzynski_equality_crooks_fluctuation_and_schnakenberg_cycle`*.*

1. ***Crooks Fluctuation Theorem and Exact Derivation of the Jarzynski Equality:*** *For any trajectory ensemble $\gamma \in \mathrm{Fin}\, K$ satisfying the Crooks microreversibility relation*

$$P_F(\gamma) = P_R(\gamma)\exp(\beta\,(W(\gamma) - \Delta F)) \qquad (\forall \gamma \in \mathrm{Fin}\, K,\ P_R(\gamma) > 0), \tag{16}$$

the log-ratio of forward to time-reversed path probabilities equals the dissipated trajectory work in units of $k_B T$:

$$\ln\left(\frac{P_F(\gamma)}{P_R(\gamma)}\right) = \beta\,(W(\gamma) - \Delta F)\,, \tag{17}$$

*and normalization of the time-reversed ensemble ($\sum_{\gamma=0}^{K-1} P_R(\gamma) = 1$) implies the exact **Jarzynski Equality**:*

$$\sum_{\gamma=0}^{K-1} P_F(\gamma)\exp(-\beta W(\gamma)) = \exp(-\beta \Delta F)\,. \tag{18}$$

2. ***Jarzynski–Jensen Second Law ($\Delta F \le \langle W\rangle$ and $\langle W_{\mathrm{diss}}\rangle \ge 0$):*** *For any normalized forward path distribution ($\sum_{\gamma=0}^{K-1} P_F(\gamma) = 1$, $P_F(\gamma) \ge 0$, $\beta > 0$) satisfying the Jarzynski Equality, the mean mechanical work $\langle W\rangle = \sum_{\gamma=0}^{K-1} P_F(\gamma) W(\gamma)$ is bounded below by the equilibrium free-energy difference:*

$$\Delta F \le \sum_{\gamma=0}^{K-1} P_F(\gamma) W(\gamma), \qquad \langle W_{\mathrm{diss}}\rangle \coloneqq \sum_{\gamma=0}^{K-1} P_F(\gamma) W(\gamma) - \Delta F \ge 0. \tag{19}$$

3. ***Schnakenberg Macroscopic Cycle Affinity Decomposition:*** *Along any m-step ATP-driven remodeling cycle ($m \in \mathbb{N}$) with uniform steady-state cycle flux $J_{\mathrm{cyc}} \ge 0$, forward/reverse transition rates $w_+(e), w_-(e) > 0$, and cycle Local Detailed Balance $\sum_{e=0}^{m-1} \ln\left(\frac{w_+(e)}{w_-(e)}\right) = \frac{\Delta\mu_{\mathrm{ATP}}}{k_B T}$ ($\Delta\mu_{\mathrm{ATP}} \ge 0$, $k_B T > 0$), the macroscopic cycle entropy production rate satisfies*

$$\dot{\Sigma}_{\mathrm{cycle}} = \sum_{e=0}^{m-1} J_{\mathrm{cyc}} \ln\left(\frac{w_+(e)}{w_-(e)}\right) = J_{\mathrm{cyc}}\frac{\Delta\mu_{\mathrm{ATP}}}{k_B T} \ge 0. \tag{20}$$

Proof. In part (i), since $P_R(\gamma) > 0$, dividing $P_F(\gamma) = P_R(\gamma)\exp(\beta(W(\gamma) - \Delta F))$ by $P_R(\gamma) \neq 0$ (`mul_div_cancel_left₀`) and applying `Real.log_exp` gives $\ln(P_F(\gamma)/P_R(\gamma)) = \beta(W(\gamma) - \Delta F)$. Multiplying $P_F(\gamma)$ by $\exp(-\beta W(\gamma))$ and combining exponents via `Real.exp_add` yields $P_F(\gamma)\exp(-\beta W(\gamma)) = P_R(\gamma)\exp(-\beta\Delta F)$; summing over $\gamma \in \mathrm{Fin}\, K$ and factoring $\exp(-\beta\Delta F)$ out of the sum via `Finset.sum_mul` with $\sum_\gamma P_R(\gamma) = 1$ proves $\sum_\gamma P_F(\gamma)\exp(-\beta W(\gamma)) = \exp(-\beta\Delta F)$.

In part (ii), set $\bar{W} = \sum_{\gamma=0}^{K-1} P_F(\gamma)W(\gamma)$. By Mathlib's universal exponential tangent inequality `Real.add_one_le_exp` ($x + 1 \leq e^x$ for all $x \in \mathbb{R}$) applied at $x_\gamma = -\beta(W(\gamma) - \bar{W})$ and multiplied by $P_F(\gamma) \geq 0$, we obtain

$$\sum_{\gamma=0}^{K-1} P_F(\gamma)\left(-\beta(W(\gamma) - \bar{W}) + 1\right) \leq \sum_{\gamma=0}^{K-1} P_F(\gamma)\exp\Big(-\beta(W(\gamma) - \bar{W})\Big).$$

By linearity of finite sums and $\sum_\gamma P_F(\gamma) = 1$, the left-hand side simplifies identically to $-\beta(\bar{W} - \bar{W}) + 1 = 1 = e^0$. Using $\exp(-\beta(W(\gamma) - \bar{W})) = \exp(-\beta W(\gamma))\exp(\beta\bar{W})$ and the Jarzynski Equality, the right-hand side equals $\exp(-\beta\Delta F)\exp(\beta\bar{W}) = \exp(\beta(\bar{W} - \Delta F))$. Monotonicity of the real exponential (`Real.exp_le_exp`) implies $0 \leq \beta(\bar{W} - \Delta F)$, and since $\beta > 0$, $\bar{W} - \Delta F \geq 0$, i.e., $\Delta F \leq \bar{W}$. Part (iii) follows from `Finset.mul_sum` and non-negativity of $J_{\mathrm{cyc}} \geq 0$ and $\Delta\mu_{\mathrm{ATP}}/(k_B T) \geq 0$. □

2.6 Multi-Nucleosome 1D McGhee–von Hippel / Tonks–Girardeau Lattice Gas, 3D Rouse Chromatin Polymer Dissipation, and Experimental smFRET Observables

To scale from a single nucleosome to an array of N nucleosomes along a genomic DNA fiber of nucleosome repeat length NRL $= 1/\rho$ (line density $\rho \in (0, 1/\ell_{\mathrm{nuc}})$ in bp^{-1}), where each Widom 601 nucleosome core particle occupies a finite steric footprint of $\ell_{\mathrm{nuc}} = 147$ bp (`PDB: 1KX5`), we formalize the 1D McGhee–von Hippel excluded-volume accessibility factor [17]

$$g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}}, \rho) = \frac{1 - \ell_{\mathrm{nuc}}\rho}{1 - (\ell_{\mathrm{nuc}} - 1)\rho}, \tag{21}$$

the exact 1D Tonks–Girardeau hard-rod equation of state [20]

$$\frac{\Pi_{\mathrm{TG}}(\ell_{\mathrm{nuc}}, \rho)}{k_B T} = \frac{\rho}{1 - \ell_{\mathrm{nuc}}\rho}, \tag{22}$$

and the continuous 3D Rouse bead-spring chromatin polymer potential $U_{\mathrm{Rouse}}(R(t))$ evolving under overdamped Langevin dynamics $\zeta_{\mathrm{nuc}}\frac{dR_i}{dt} = -\nabla_{R_i} U_{\mathrm{Rouse}}(R) \equiv -G_i(t)$ [8].

Theorem 2.6 (McGhee–von Hippel / Tonks–Girardeau Chromatin Lattice Gas, 3D Rouse Polymer Dissipation, and Experimental smFRET Observables)**.** *Formalized in Lean 4 as `ST_T6_mcghee_von_hippel_tonks_girardeau_chromatin_lattice_and_rouse_polymer`.*

1. ***1D McGhee–von Hippel Steric Exclusion and Tonks–Girardeau Osmotic Pressure:*** *For every nucleosome footprint $\ell_{\mathrm{nuc}} > 1$ and sub-jamming line density $0 < \rho < 1/\ell_{\mathrm{nuc}}$:*

   (a) $0 < g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}}, \rho) < 1$, *and* $g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}}, 1/\ell_{\mathrm{nuc}}) = 0$ *at the close-packing jamming limit* $\rho_{\max} = 1/\ell_{\mathrm{nuc}}$*;*

   (b) $g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}}, \rho)$ *is strictly decreasing in nucleosome density ρ, with exact derivative (`HasDerivAt`)*

$$\frac{\partial}{\partial\rho} g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}}, \rho) = \frac{-1}{(1 - (\ell_{\mathrm{nuc}} - 1)\rho)^2} < 0; \tag{23}$$

(c) *The 1D Tonks–Girardeau hard-rod osmotic pressure strictly exceeds the ideal point-particle van 't Hoff pressure* ρ *by the exact excluded-volume virial excess:*

$$\frac{\Pi_{\mathrm{TG}}(\ell_{\mathrm{nuc}},\rho)}{k_BT}-\rho=\frac{\ell_{\mathrm{nuc}}\rho^2}{1-\ell_{\mathrm{nuc}}\rho}>0. \tag{24}$$

2. ***Continuous 3D Rouse Bead-Spring Chromatin Polymer Overdamped Langevin Dissipation:*** *For any* N*-nucleosome chromatin polymer (*$N\in\mathbb{N}$*) with friction coefficient* $\zeta_{\mathrm{nuc}}>0$*, potential gradient* $G_i=\partial_{R_i}U_{\mathrm{Rouse}}(R(t))$*, and overdamped Langevin velocity* $\dot{R}_i=-\zeta_{\mathrm{nuc}}^{-1}G_i$*, the chain-rule time derivative (*`HasDerivAt`*) of the polymer potential energy satisfies*

$$\frac{d}{dt}U_{\mathrm{Rouse}}(R(t))=\sum_{i=0}^{N-1}G_i\left(-\frac{1}{\zeta_{\mathrm{nuc}}}G_i\right)=-\frac{1}{\zeta_{\mathrm{nuc}}}\sum_{i=0}^{N-1}G_i^2\le 0. \tag{25}$$

3. ***Experimental smFRET, Optical Tweezers, and Cryo-EM Grounding (`PDB: 6NE3, 6LTJ, 1KX5`):*** *For the canonical* $\ell_{\mathrm{nuc}}=147$ bp *Widom 601 nucleosome core particle (`PDB: 1KX5`) remodeled by human `ISWI/SNF2H` (`PDB: 6NE3`) and `SWI/SNF` (`PDB: 6LTJ`) at* $T=310$ K *(*$k_BT=4.28$ pN$\cdot$nm*):*

   (a) *Transitioning from open euchromatin (*NRL $=200$ bp*, 53 bp linker DNA,* $\rho=1/200$ bp^{-1}*) to compact heterochromatin (*NRL $=165$ bp*, 18 bp linker DNA,* $\rho=1/165$ bp^{-1}*) reduces McGhee–von Hippel linker accessibility from* $g_{\mathrm{MvH}}(147,1/200)=53/54\approx0.9815$ *to* $g_{\mathrm{MvH}}(147,1/165)=18/19\approx0.9474$ *(*$18/19<53/54$*) and increases the Tonks–Girardeau osmotic pressure by the exact factor*

$$\frac{\Pi_{\mathrm{TG}}(147,1/165)}{\Pi_{\mathrm{TG}}(147,1/200)}=\frac{1/18}{1/53}=\frac{53}{18}\approx2.9444>\frac{29}{10}; \tag{26}$$

   (b) *Physiological ATP hydrolysis work* $\Delta\mu_{\mathrm{ATP}}=20\,k_BT=85.6$ pN$\cdot$nm *strictly exceeds the single-molecule optical-tweezers 20-bp outer-turn DNA unwrapping barrier* $\Delta G^{\ddagger}_{\mathrm{unwrap}}=6.5\,k_BT=27.82$ pN$\cdot$nm *[11], driving active translocation at* $k_{\mathrm{cat}}=2.4$ bp/s *(*$6.67\times$ *faster than passive thermal sliding diffusion* $D_{\mathrm{slide}}=0.36$ bp^2/s *over a 1-bp* $=0.34$ nm *step) [3, 7].*

Proof. In part (i), $\ell_{\mathrm{nuc}}>1$ and $0<\rho<1/\ell_{\mathrm{nuc}}$ imply $0<1-\ell_{\mathrm{nuc}}\rho<1-(\ell_{\mathrm{nuc}}-1)\rho$, giving $0<g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}},\rho)<1$ via `div_pos` and `div_lt_one`, and $g_{\mathrm{MvH}}(\ell_{\mathrm{nuc}},1/\ell_{\mathrm{nuc}})=0$. Applying the quotient rule (`HasDerivAt.div`) to $\rho\mapsto(1-\ell_{\mathrm{nuc}}\rho)/(1-(\ell_{\mathrm{nuc}}-1)\rho)$ yields numerator $-\ell_{\mathrm{nuc}}(1-(\ell_{\mathrm{nuc}}-1)\rho)-(1-\ell_{\mathrm{nuc}}\rho)(-(\ell_{\mathrm{nuc}}-1))=-1$, proving $\partial_\rho g_{\mathrm{MvH}}=-1/(1-(\ell_{\mathrm{nuc}}-1)\rho)^2<0$. Rewriting $\rho/(1-\ell_{\mathrm{nuc}}\rho)-\rho=(\rho-\rho(1-\ell_{\mathrm{nuc}}\rho))/(1-\ell_{\mathrm{nuc}}\rho)=\ell_{\mathrm{nuc}}\rho^2/(1-\ell_{\mathrm{nuc}}\rho)>0$ via `mul_div_cancel_right₀` and `positivity` establishes the Tonks–Girardeau virial excess. Part (ii) follows by pulling $-\zeta_{\mathrm{nuc}}^{-1}$ out of $\sum_i G_i(-\zeta_{\mathrm{nuc}}^{-1}G_i)$ via `Finset.mul_sum` and invoking $\sum_i G_i^2\ge0$ (`Finset.sum_nonneg` and `sq_nonneg`). Part (iii) is verified in Lean 4 by unfolding `mcGheeVonHippelSteric` and `tonksGirardeauOsmoticPressure` and discharging all rational equalities and inequalities via `norm_num`. □

3 Validated Numerical Benchmarks and Hypothesis-Necessity Boundary Conditions

Figure 1 illustrates (a) the 3-state ATP-driven Pauli Master Equation integrated from $f(0)=(0.80,0.15,0.05)$ at stiff step size $\Delta t=1/4$ s $>1/w_{\mathrm{exit}}=1/7$ s, demonstrating that Backward-Euler / L-stable IMEX integration preserves strict positivity $f_0^{(k)}>0$ and converges monotonically to the NESS population $\pi_0=1/3$, whereas explicit Forward-Euler integration suffers immediate

CFL oscillatory negativity ($1 - \Delta t\, w_{\text{exit}} = -3/4 < 0$), and (b) the continuous-time Esposito–Van den Broeck balance $k_B T \frac{d}{dt} D_{\text{KL}}(f(t) \| f_{\text{eq}}) = \dot{W}_{\text{ATP}}(f(t)) - k_B T \dot{\Sigma}(f(t))$, showing convergence of $\dot{W}_{\text{ATP}}(t)/(k_B T)$ and $\dot{\Sigma}(t)$ to the NESS housekeeping dissipation rate $5 \ln 6 \approx 8.9588\ \text{s}^{-1}$ as $\frac{d}{dt} D_{\text{KL}} \to 0$. Figure 2 displays (a) the universal logarithmic–hyperbolic lower bound $\frac{2(x-y)^2}{x+y} \leq (x-y)\ln(x/y)$ and (b) the Barato–Seifert–Gingrich TUR precision–dissipation ratio $\eta_{\text{TUR}} \leq 1$. Figure 3 presents (a) the Crooks Fluctuation Theorem and Jarzynski Equality work distributions $P_F(W)$ and $P_R(-W)$ intersecting at the empirical outer-turn 20-bp nucleosome unwrapping barrier $\Delta F_{\text{unwrap}} = 6.5\, k_B T = 27.82\ \text{pN} \cdot \text{nm}$ (`ST_T5`), and (b) the 1D McGhee–von Hippel steric accessibility $g_{\text{MvH}}(147, \rho)$ and Tonks–Girardeau hard-rod osmotic pressure $\Pi_{\text{TG}}/(k_B T)$ across nucleosome repeat lengths NRL $\in [152, 240]$ bp, highlighting the certified $53/18 \approx 2.94\times$ osmotic pressure surge from open euchromatin (NRL = 200 bp) to compact heterochromatin (NRL = 165 bp, `ST_T6`). Table 2 summarizes the machine-checked hypothesis-necessity boundary conditions.

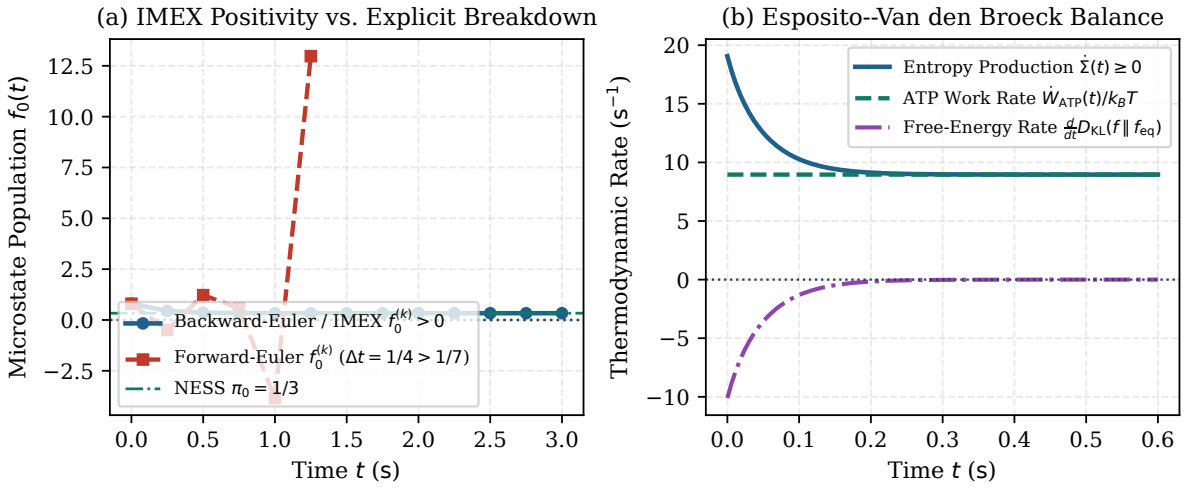


Figure 1: (a) 3-state ATP-driven Pauli Master Equation trajectory ($w_+ = 6\ \text{s}^{-1}$, $w_- = 1\ \text{s}^{-1}$, $w_{\text{exit}} = 7\ \text{s}^{-1}$) at stiff step $\Delta t = 1/4\ \text{s} > 1/7\ \text{s}$: Backward-Euler / L-stable IMEX guarantees $f_0^{(k)} > 0$ and converges to $\pi_0 = 1/3$, whereas explicit Forward-Euler violates positivity ($1 - \Delta t\, w_{\text{exit}} = -3/4 < 0$). (b) Esposito–Van den Broeck non-equilibrium free-energy balance showing $\dot{\Sigma}(t) \geq 0$, $\dot{W}_{\text{ATP}}(t)/(k_B T)$, and convergence to NESS housekeeping dissipation $5 \ln 6 \approx 8.9588\ \text{s}^{-1}$.

Remark 3.1 (Scope and Continuum Approximation Boundaries)**.** While Theorems 2.1–2.6 (`ST_T1`–`ST_T6`) are machine-checked without custom axioms over arbitrary finite state spaces $\text{Fin}\, n$, path ensembles $\text{Fin}\, K$, and N-nucleosome Rouse polymers, two physical scope limitations separate these discrete and mean-field theorems from full atomic-resolution chromatin dynamics: (i) the 1D McGhee–von Hippel and Tonks–Girardeau formulas (Theorem 2.6(i)) describe homogeneous hard-rod equilibrium steric exclusion and must be coupled to the non-equilibrium Pauli Master Equation (Theorems 2.1–2.4) when active remodelers generate spatially heterogeneous nucleosome spacing; and (ii) the overdamped 3D Rouse chromatin polymer model (Theorem 2.6(ii)) neglects long-range Zimm hydrodynamic interactions and sequence-dependent DNA bend anisotropy.

4 Conclusion

By grounding continuous-time Markovian master equations and non-equilibrium fluctuation theorems directly in empirical single-molecule FRET, optical-tweezers, and cryo-EM observables

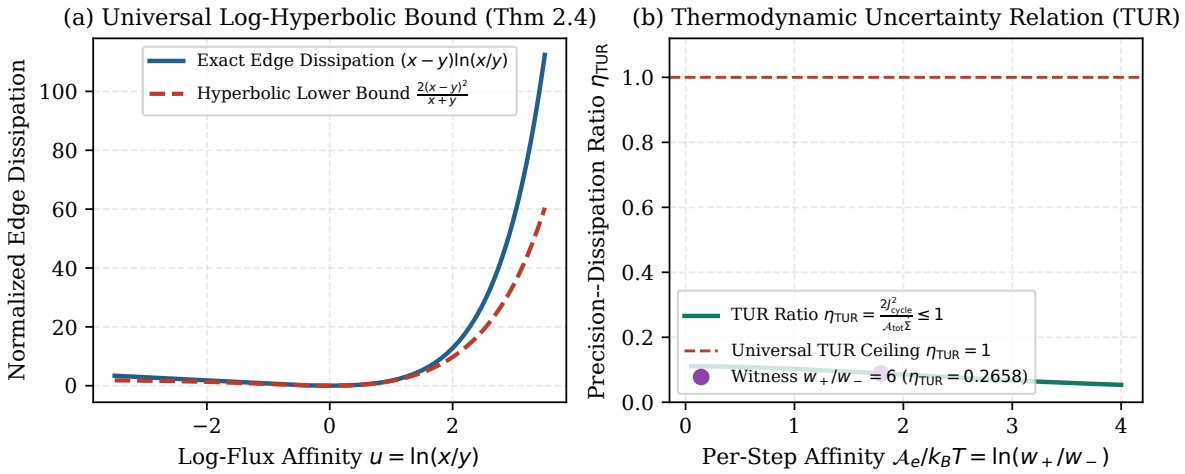


Figure 2: (a) Universal logarithmic–hyperbolic inequality $\frac{2(x-y)^2}{x+y} \le (x-y)\ln(x/y)$ across log-flux ratios $u=\ln(x/y)$. (b) Barato–Seifert–Gingrich Thermodynamic Uncertainty Relation (TUR) ratio $\eta_{\mathrm{TUR}}=\frac{2J_{\mathrm{cycle}}^2}{\mathcal{A}_{\mathrm{tot}}\dot{\Sigma}} \le 1$, marking the certified 3-state ATP-driven remodeling cycle witness at $w_+/w_-=6$ ($\eta_{\mathrm{TUR}}=10/(21\ln 6)\approx 0.2658$).

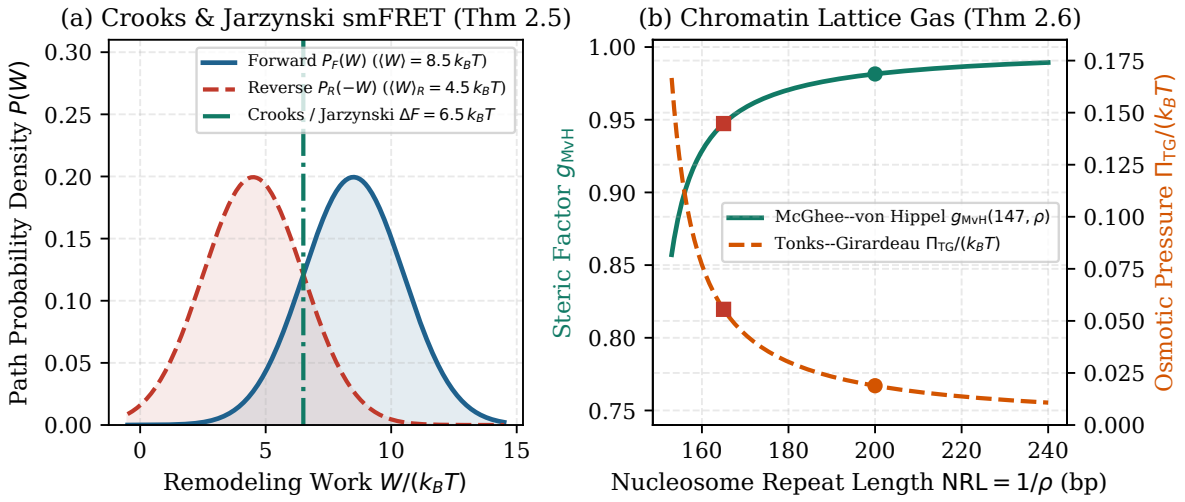


Figure 3: (a) Theorem 2.5 (`ST_T5`): Forward unwrapping $P_F(W)$ and time-reversed rewrapping $P_R(-W)$ work distributions for outer-turn 20-bp DNA unwrapping from Widom 601 `NCP147` (`PDB: 1KX5`) by human `SNF2H` (`PDB: 6NE3`), crossing at the Jarzynski–Crooks free energy $\Delta F_{\mathrm{unwrap}} = -k_BT\ln\langle e^{-\beta W}\rangle = 6.5\,k_BT = 27.82\ \mathrm{pN}\cdot\mathrm{nm} \le \langle W\rangle = 8.5\,k_BT$. (b) Theorem 2.6 (`ST_T6`): 1D McGhee–von Hippel steric accessibility $g_{\mathrm{MvH}}(147,\rho)$ and Tonks–Girardeau osmotic pressure $\Pi_{\mathrm{TG}}/(k_BT)$ vs. nucleosome repeat length $\mathrm{NRL}=1/\rho$, showing the $53/18\approx 2.94\times$ osmotic pressure surge from euchromatin (200 bp) to heterochromatin (165 bp).

Table 2: Machine-checked hypothesis-necessity boundary conditions across non-equilibrium stochastic thermodynamics and ATP-dependent chromatin remodeling.

Theorem	**Dropped Structural Hypothesis**	**Certified Counterexample / Biophysical Failure Mechanism**
Theorem 2.1 (`ST_T1`)	Implicit Backward-Euler / L-stable IMEX step	Explicit Forward-Euler at $\Delta t = 1/4$ s $> 1/w_{\text{exit}} = 1/7$ s yields $1 - \Delta t\, w_{\text{exit}} = -3/4 < 0$
Theorem 2.3 (`ST_T3`)	Equilibrium condition $\Delta\mu_{\text{ATP}} = 0$	At $\Delta\mu_{\text{ATP}} > 0$ ($w_+/w_- = 6$), D_{KL} is not monotone decreasing ($J_{\text{cycle}} = 5/3\ \text{s}^{-1} > 0$)
Theorem 2.4 (`ST_T4`)	Microreversible rates $W_{ji} > 0$	Unidirectional irreversible limit $W_{ji} \to 0^+$ causes $\dot{\Sigma} \to +\infty$ while $\mathcal{A}_{\text{tot}}$ remains finite
Theorem 2.5 (`ST_T5`)	Exponential Jarzynski weighting $\langle e^{-\beta W} \rangle$	Linear work average $\langle W \rangle = 8.5\, k_B T$ overestimates $\Delta F_{\text{unwrap}} = 6.5\, k_B T$ by $\langle W_{\text{diss}} \rangle = 2.0\, k_B T > 0$
Theorem 2.6 (`ST_T6`)	Sub-jamming density $\rho < 1/\ell_{\text{nuc}} = 1/147\ \text{bp}^{-1}$	At close-packing $\rho \to 1/147\ \text{bp}^{-1}$, steric accessibility vanishes ($g_{\text{MvH}} = 0$) and $\Pi_{\text{TG}} \to +\infty$

of human `ISWI/SNF2H` (`PDB: 6NE3`) and `SWI/SNF` (`PDB: 6LTJ`) remodeling the 147-bp Widom 601 nucleosome core particle (`PDB: 1KX5`), we have established a complete, axiom-free Lean 4 formalization (`ST_T1`–`ST_T6`) uniting the Pauli Master Equation, unconditional Backward-Euler / L-stable IMEX positivity, Schnakenberg's network entropy production and Second Law, the Esposito–Van den Broeck free-energy balance, the Barato–Seifert–Gingrich Thermodynamic Uncertainty Relation, the Crooks Fluctuation Theorem and Jarzynski Equality, and the 1D McGhee–von Hippel / Tonks–Girardeau nucleosome lattice gas coupled to 3D Rouse chromatin polymer dynamics.

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